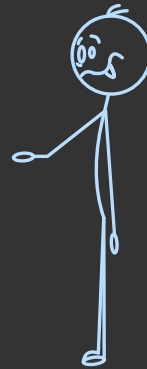


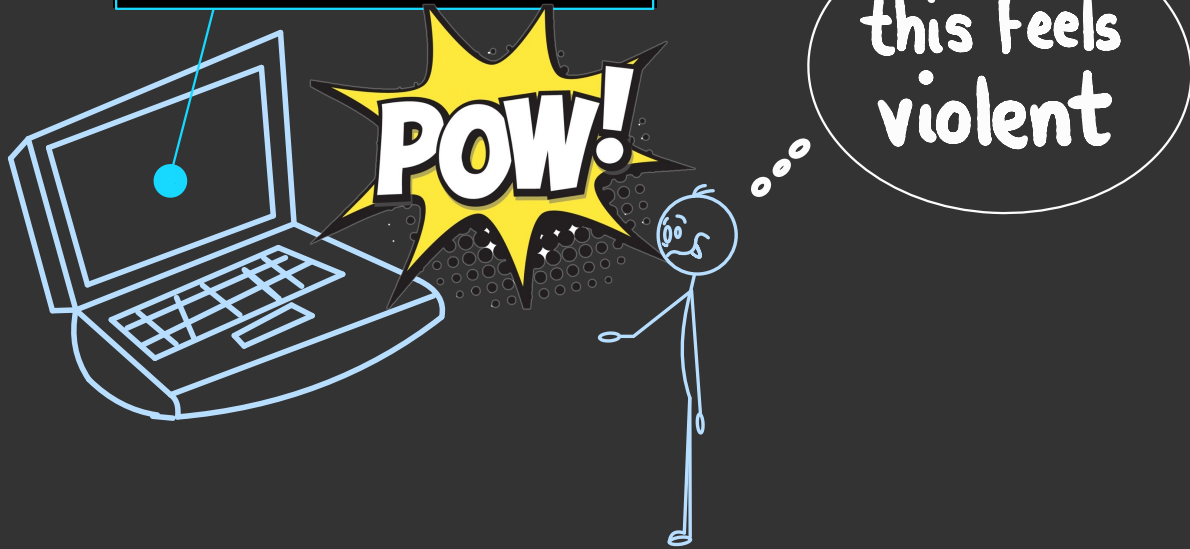
lambda expressions

CSCI
134

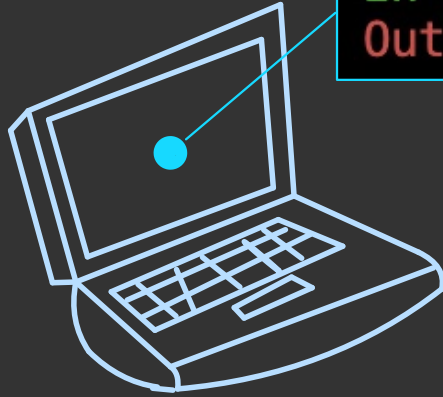
```
In [1]: pow(3, 2)  
Out[1]: 9
```



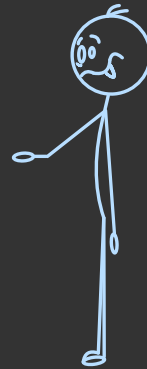
```
In [1]: pow(3, 2)  
Out[1]: 9
```



what if we wanted to rename
the **pow** function?



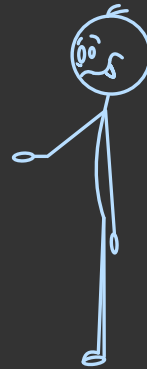
```
In [1]: power = pow  
  
In [2]: power(3, 2)  
Out[2]: 9
```



variable = expression

since functions are expressions, we could
just use an **assignment** statement

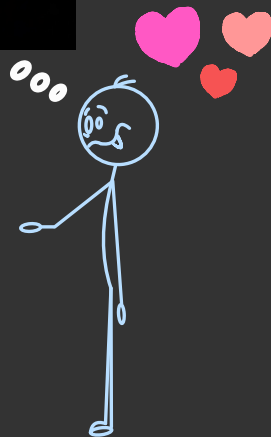
```
In [1]: pow(3, 2)  
Out[1]: 9
```



base, then
exponent?
ew.

but what if we want an **alternative**
version that swaps the argument order?

```
In [3]: power(3, 2)  
Out[3]: 8
```



we need a way to create
new functions

one way to create a new function

```
In [1]: power = lambda exp, base: pow(base, exp)
```

```
In [2]: pow(3, 2)
```

```
Out[2]: 9
```

```
In [3]: power(3, 2)
```

```
Out[3]: 8
```

one way to create a new function

```
lambda exp, base: pow(base, exp)
```

arguments

(comma-separated)

return value

(what the function computes)

one way to create a new function

Lambda

arguments

(comma-separated)

:

return value

(what the function computes)

exercise: create a function
that **doubles** a number

```
In [1]: double = 
```

?

```
In [2]: double(3)
```

```
Out[2]: 6
```

```
In [3]: double(20)
```

```
Out[3]: 40
```

remember!

lambda

arguments

(comma-separated)

:

return value

(what the function computes)

exercise: create a function
that **doubles** a number

```
In [1]: double = lambda x : 2*x
```

```
In [2]: double(3)
```

```
Out[2]: 6
```

```
In [3]: double(20)
```

```
Out[3]: 40
```

remember!

lambda

arguments

(comma-separated)

:

return value

(what the function computes)

exercise: create a function that
computes the **length difference** of two strs

```
In [1]: diff =
```

?

```
In [2]: diff("hello", "there")
```

```
Out[2]: 0
```

```
In [3]: diff("rohit", "bhattacharya")
```

```
Out[3]: 7
```

```
In [4]: diff("katherine", "keith")
```

```
Out[4]: 4
```

remember!

lambda

arguments

(comma-separated)

:

return value

(what the function computes)

exercise: create a function that
computes the **length difference** of two strs

```
In [1]: diff = lambda s1, s2 : abs(len(s1) - len(s2))
```

```
In [2]: diff("hello", "there")  
Out[2]: 0
```

```
In [3]: diff("rohit", "bhattacharya")  
Out[3]: 7
```

```
In [4]: diff("katherine", "keith")  
Out[4]: 4
```

remember!

lambda

arguments
(comma-separated)

:

return value
(what the function computes)